New challenges for an NSBI measurement of κ_λ in HH$\to b\bar{b}\tau\tau$, relative to the first NSBI measurement (ATLAS off-shell $H \to ZZ \to 4\ell$)

working note — draft for discussion

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Scope & one-line conclusion

This compares the systematics and background handling of the **first full NSBI measurement** (ATLAS off-shell $H \to ZZ \to 4\ell$; note HIGG-2023-01 = arXiv:2412.01600, Sandesara *et al.*) with what a κ_λ **measurement in HH**$\to b\bar{b}\tau\tau$ must do (baselined on CMS AN-18-121, Amendola *et al.*). The NSBI *skeleton* transfers directly; the new difficulty is almost entirely in the *inputs*: $b\bar{b}\tau\tau$ has a far messier final state, several **data-driven backgrounds with no per-event MC density** (QCD, jet$\to\tau$ fakes), an **irreducible same-final-state single-Higgs background**, an order of magnitude more (and larger) **experimental systematics that deform the very observable (m_{HH}) carrying the κ_λ signal**, and a κ_λ**-dependent theory uncertainty**. None of these are NSBI-specific — they hit any method — but several sit awkwardly inside a per-event density-ratio framework in ways the clean, all-MC off-shell analysis never faced. Page references to both notes are given inline.

Studies to do

Concrete studies to de-risk and execute this measurement, grouped by theme. Priority: **[P0]** blocking / do-first, **[P1]** important, **[P2]** nice-to-have; effort (S <1 wk, M weeks, L months). Trailing [§n] point to the relevant numbered section below.

Critical path

Statistics-first: port the JAX+IMINUIT fit (#11) and fix P_{ref} (#17), then build the in-situ-CR likelihood (#12) and the canonical ABCDnn flow with ExtendedABCD yields (#1,#2) in parallel; only *then* run Asimov closure (#13) and Neyman coverage (#14). The JES$\times\tau$ES/THU factorisation (#5), the coherent τES$\to$SVfit chain (#6), and the feature-ablation degeneracy study (#18) must land *before* coverage so the morphing is trustworthy; honest GP morphing-uncertainty (#9) is gated behind the MC-stat ensemble (#15). Multi-channel (#23), the k_{2V} tail (#21), POI-correlations (#22), and pipeline/pretraining (#25,#26) proceed as P1/P2 follow-ups.

Data-driven backgrounds

1. **[P0]** (L) **Canonical ABCDnn flow for QCD & jet$\to\tau$ fakes** (yield/shape split, three-piece uncertainty). MMD-trained NAF (continuous (N_j, N_b) conditioning, identity warm-up) morphs MC $\to$ the data-driven CR target; yield from ExtendedABCD/fake-factors, flow $\to$ per-event ratio $r_{\text{QCD}}(x)$ via a flow-vs-reference classifier; port the three four-top nuisances as per-event morphing; sweep target dim 2D$\to$5D$\to$10D. **Done:** VR closure $|1-\langle f\rangle| < 0.05$, RMS < 0.15; $r(x) \geq 0$ integrating to the ABCD yield; max stable dimension documented. [§4]

2. **[P0]** (M) **ExtendedABCD vs standard-ABCD yield closure for $b\bar{b}\tau\tau$ QCD.** Standard ABCD failed $\sim$18–19% at high N_j in the four-top analysis and the QCD-norm NP is already large; test the extended estimator's closure across 6–8 orthogonal SRs binned in (N_j, N_b). **Done:** bias $< 7\%$, RMS $< 12\%$ per multiplicity bin; a safe-vs-extended region map. [§4]

3. **[P1]** (S) **Encode the one-sided QCD shape NP into smooth morphing.** CMS uses one alternative template for both up/down; test mirrored / one-sided-exponential / half-Gaussian encodings so the poly-exp morphing of $r_{\mathrm{QCD}}(x)$ does not oscillate. **Done:** matches the CMS shape-impact rank ± 2, no oscillation, $< 2\%$ κ_λ bias. [§4]

4. **[P1]** (L) **Quantify the binned-QCD/fakes vs fully-unbinned cost.** Asimov ($\kappa_\lambda = 0, 1, 5$): full unbinned vs MC-only NSBI with QCD/fakes as a fixed binned template; also bin the NSBI score at 5/10/20/50 to check convergence. **Done:** width inflation $< 15\%$, bias $< 0.3\sigma$, monotonic convergence; a bin-vs-unbin fallback decision tree. [§4, §7]

Systematics & learned morphing

5. **[P0]** (M) **Validate factorisation & cross-terms: JES$\times\tau$ES and THU$_{HH}\times\kappa_\lambda$.** Both JES and τES feed m_{HH} (a cross-term absent in clean 4ℓ) and THU$_{HH}$ is POI-coupled. Test $r_{\mathrm{comb}}/\prod_m r_m$ (median, KL) on held-out $\pm 1\sigma$ MC; compare naive-lnN vs κ_λ-coupled THU in Asimov. **Done:** factorisation residual 0.98–1.02, KL < 0.02; add a correction-NN if $> 2\%$. [§5]

6. **[P0]** (S) **Validate the coherent τES$\to$MET$\to$SVfit$\to m_{HH}$ recompute.** The weight-builder must shift τ energy at object level and recompute MET+SVfit+m_{HH} end-to-end, not shift m_{HH} directly. **Done:** matches the coherent truth to $< 2\%$ per-event MAE; the shortcut bias quantified. [§3, §5]

7. **[P1]** (M) **Validate poly-exp interpolation (κ_λ scan + $\pm 1\sigma$ anchors) under limited MC.** Intermediate κ_λ (0.5, 1.5) vs reweighted truth; downsample $\pm 1\sigma$ samples 100$\to$10%. **Done:** MAE $< 5\%$ down to $\sim$50% of MC, no gradient oscillation; minimum $\pm 1\sigma$ sample size to request from generators. [§1, §5]

8. **[P1]** (M) **Build & profile a CARL/DCTR per-event parton-shower/generator nuisance** (the NSBI upside). Train the per-event weight Pythia$\leftrightarrow$Herwig / ISR/FSR / VBF dipole-recoil; profile as one Gaussian NP vs the binned envelope. **Done:** KS $p > 0.1$, interval $\leq$ envelope with correct coverage, generator closure absorbed $< 2\sigma$. [§5]

9. **[P1]** (M) **Honest GP/ensemble-spread morphing-uncertainty with coverage calibration.** Fit a GP to per-event $r_{\mathrm{true}} - r_{\mathrm{pred}}$ residuals as a phase-space-dependent band (gated behind #15). **Done:** 65–70% coverage, calibration slope $1 \pm 10\%$, Spearman $\rho > 0.6$, narrower than a global band. [§5, §7]

10. [P2] (M) **Absorb b-tag/τ-ID SFs into the ratio, calibrated in situ** ($Z \to b\bar{b}$ / VZ($c\bar{c}$)). Demote the POG SF program to a cross-check and shrink the NN count. **Done:** in-situ SF within 30% of the POG prior, κ_λ not degraded, $|\Delta\mathrm{pull}(0 \text{ vs } 5)| < 0.5\sigma$. [§5]

Statistics, fit & coverage

11. **[P0]** (M) **Port the jax+iminuit profile fit with the $\{\kappa_\lambda{}^2, \kappa_\lambda, 1\}$ basis** (root dependency). Replicate ATLAS Sec. 5: P_{ref}, the signal/interference/background mixture with κ_λ-dependent rate, $\pm 1\sigma$ ratio classifiers + poly-exp, profile fit with Gaussian-constrained NPs; cross-check vs a binned Asimov. **Done:** converges, Hesse invertible, parabolic NLL; unbinned-vs-binned width $< 5\%$. [§1, §6]

12. **[P0]** (M) **Represent the in-situ control regions (18 DY SFs, $t\bar{t}$-SF, QCD B/D) inside the unbinned likelihood.** Add Poisson CR auxiliary terms sharing the same NPs; compare SR-only-free-NP vs SR+explicit CRs. **Done:** reproduce the CMS in-situ SFs within 10–20%, no double-counting, corr(SF,κ_λ)< 0.3. [§4, §6]

13. **[P0]** (M) **Asimov & closure: morphing bias under injected shifts and generator/Delphes mismatch.** Inject known $\pm1\sigma$ on JES/τES/QCD/MC-stat and re-fit; fit data from chain X with a model trained on Y (Pythia$\leftrightarrow$Herwig; Delphes$\leftrightarrow$full-sim). **Done:** recover κ_λ to < 0.05, pulls $\sim N(0,1)$, sim-closure bias < 0.1 (flag full-sim retraining if $> 10\%$). [§5, §6]

14. **[P0]** (M) **Neyman construction & coverage validation on MC-truth toys.** 1000+ toys; build the belt at $\kappa_\lambda \in \{-1, 0, 1, 3, 5\}$. **Done:** empirical coverage $\geq 67\%$, $|\text{bias}| < 0.1$ at SM; any failure traced to a specific NP/component. [§6]

15. **[P1]** (M) **MC-stat ensemble-spread nuisance: design, size, coverage, Barlow–Beeston orthogonality.** Compare ensemble strategies (seeds / k-fold / bootstrap / m_{HH}-stratified) for the per-event ratio. **Done:** C2ST $p > 0.1$ on holdout, NP width ≈ 1 (BB-valid), κ_λ widens 5–15% when unfrozen, corr < 0.3 with physics NPs. [§5, §6]

16. **[P1]** (S) **Interference-aware impact definition + binned-NSBI fallback.** Implicit-function-theorem impact at the minimum vs finite-difference and the CMS rank; histogram the per-event log-ratio (5/10/20 bins) as an exact, learning-free cross-check. **Done:** matches the CMS rank ±2; binned constraint $\geq 80\%$ of unbinned, monotonic convergence. [§6, §7]

Physics & POI

17. **[P0]** (S) **Define & validate the reference hypothesis P_{ref}.** Compare SM / κ_λ-flat-mixture / pure-background / dedicated references; measure C2ST closure and how the $\{{\kappa_\lambda}^2, \kappa_\lambda, 1\}$ reconstruction degrades with $|\kappa_\lambda - \kappa_{\lambda\text{ref}}|$, including the thin high-m_{HH} k_{2V} tail. **Done:** C2ST $p > 0.05$ across the scan, $|\log r| < 10$, $< 5\%$ basis bias at extremes. [§1]

18. **[P0]** (M) **Feature-ablation degeneracy-breaking: $\rho(\kappa_\lambda$,JES/τES$)$ and VIF vs feature schedule.** $m_{HH}\to$10-feature$\to$ +constituents$\to$ +SVfit covariance$\to$ +resolution handles; ablate to find the critical features and the asymptotic ρ floor. **Done:** reproduce $\rho\, 1.000 \to 0.646$; a minimal set with $\rho < 0.7$, VIF< 1.5; settle whether constituents/SVfit push $\rho < 0.60$ or it plateaus. [§2, §5]

19. **[P1]** (S) **Source the κ_λ-differential THU$_{HH}$ theory band.** Compile the scale/PDF/m_t uncertainty vs $\kappa_\lambda \in [-2, 5]$ from the LHC HH WG / Powheg-NLO and parametrise it as a POI-coupled weight. **Done:** a smooth band reproducing $+4/-18\%$ at the SM, ready for the factorisation study (#5). [§2, §5]

20. **[P1]** (M) **Validate the LO$\to$NLO κ_λ-reweighting that generates the $R(x|\kappa_\lambda)$ sources.** Direct-NLO vs 2D-LO-reweight vs dedicated-NLO-reweight; propagate the difference through an Asimov fit. **Done:** NLO-reweight reproduces direct to $< 5\%$/bin and $< 0.1\sigma$ κ_λ bias; the LO-only cost quantified. [§2]

21. **[P1]** (M) **Measure k_{2V} sensitivity in the high-m_{HH} VBF tail via unbinned NSBI.** VBF-enriched ($M_{jj} > 400\,\text{GeV}$), k_{2V}-morphed MC; Asimov-fit binned vs unbinned vs hybrid; cross-validate the tail (train low-m_{HH}, test high). **Done:** NSBI k_{2V} error 20–40% smaller, tail-extrapolation bias $< 10\%$, a minimum viable bin granularity. [§2]

22. **[P1]** (L) **Map the rate-broken POI correlations ($\kappa_t \leftrightarrow \kappa_\lambda$, $k_V \leftrightarrow k_{2V}$) and confirm the scope caveat.** Multi-POI grid; profile-likelihood correlation contours; attribute information to shape vs rate. **Done:** a POI correlation matrix; NSBI lowers $\rho(\kappa_\lambda, k_{2V})$ but does *not* improve the rate-broken pairs. [§2, §7]

23. **[P1]** (M) **Build the multi-channel + boosted simultaneous unbinned likelihood.** Per-channel vs joint morphing for the three $\tau\tau$ channels + boosted, with the CMS year/channel NP-correlation scheme. **Done:** combined κ_λ width within 5% of the best combination; no extra $> 0.5\sigma$ pulls. [§1, §3]

ML, samples & pipeline

24. **[P1]** (M) **Low-signal robustness: morphing calibration & uncertainty honesty at $\mathcal{O}$(10–100) signal events.** Downsample the $\pm1\sigma$ sets and effective signal yield (50/100/250) and retrain per

level. **Done:** C2ST $p > 0.05$ at the $\pm 1\sigma$ floor, ensemble spread $< 2\times$ the high-stat value; flag the binned fallback if it fails. [§5, §7]

25. **[P2]** (L) **Reproducible Perlmutter pipeline for the ∼2000-NN morphing ensemble.** Containerise Sophon+JAX+IMINUIT, a training manifest, resume-on-failure; benchmark per-NP classifiers vs a single conditional morphing network. **Done:** 500 NNs in < 4 h (∼1 GPU-day), $> 99\%$ resume, bit-identical reruns; the per-NP-vs-conditional trade-off documented. [general]
26. **[P2]** (M) **Pretraining follow-up: signal-vs-$t\bar{t}$ foundation-model benchmark vs scratch** (confirmatory). Scratch vs $t\bar{t}$-kinematics-pretrain vs HH-kinematics-pretrain; F-score $\kappa_\lambda = 0$ vs 5 by m_{HH} region. **Done:** signal-vs-$t\bar{t}$ pretraining closes $\geq 30\%$ of the FM deficit, else confirm scratch remains the baseline. [established null]

1 The two measurements and the shared NSBI skeleton

ATLAS off-shell [ATLAS p. 1,5]: measures the off-shell signal strength $\mu_{\text{off-shell}}$ (hence the Higgs width) in $gg \to (H^* \to) ZZ \to 4\ell$, using $139\,\text{fb}^{-1}$. The physics is a *coupling that morphs an interference shape* (triangle–box in $ggZZ$). **CMS HH→ $b\bar{b}\tau\tau$** [CMS p. 3,4]: searches for non-resonant HH via ggF+VBF in $b\bar{b}\tau\tau$ ($\mathcal{B} = 7.3\%$), $137.6\,\text{fb}^{-1}$, three $\tau\tau$ channels ($\tau_e\tau_h, \tau_\mu\tau_h, \tau_h\tau_h$, plus boosted), extracting κ_λ (and k_{2V}, k_V). $\sigma^{HH}_{ggF} = 31.05^{+2.2}_{-5.0}\%$ fb, $\sigma^{HH}_{VBF} = 1.726$ fb [CMS p. 4].
What transfers (the skeleton). The ATLAS recipe (Sec. 5) is reusable wholesale: a reference-hypothesis P_{ref} so all NNs estimate ratios to a fixed point; a search-oriented *mixture model* for signal/interference/background; nuisance parameters in both the rate and the per-event density; a profile-likelihood test statistic in a custom JAX+IMINUIT fit; and an interference-aware impact definition [ATLAS p. 89,90]. The off-shell interference structure $(\mu - \sqrt{\mu})\,S + \sqrt{\mu}\,\text{SBI} + (1 - \sqrt{\mu})\,B$ [ATLAS p. 81] maps onto the κ_λ morphing basis $\{\kappa_\lambda^2, \kappa_\lambda, 1\}$ (since $gg \to HH = \kappa_\lambda \cdot \text{triangle} + \text{box}$). *The challenges below are about the inputs to this skeleton, not the skeleton itself.*

2 Physics / parameter-of-interest challenges

- **The POI lives in a shape, and that shape is m_{HH}.** κ_λ sensitivity is concentrated in the m_{HH} spectrum (triangle–box interference), strongest in the resolved regime $m_{HH} \in [250, 400]$ GeV. CMS encodes the κ_λ dependence by *reweighting* the signal MC — a 2D ($m_{HH}, |\cos\theta^*|$) LO reweighting and a dedicated NLO reweighting for the final extraction [CMS p. 6]. In NSBI this becomes the $R(x|\kappa_\lambda)$ factor.
- **Multiple correlated POIs / EFT directions.** Beyond κ_λ the analysis also probes k_{2V}, k_V and HEFT operators (c_2, c_{2g}, c_g) [CMS p. 4]; the off-shell measurement is effectively one POI. More POIs ⇒ a higher-dimensional morphing.
- **k_{2V} sensitivity lives in the high-m_{HH} VBF tail — where binning hurts most.** The $HHVV$ contact term grows with energy unless $k_{2V} = 1$, so VBF-HH k_{2V} sensitivity is concentrated in the high-m_{HH} tail; current analyses use only ∼4–5 bins there. This is precisely the regime where coarse binning discards the most information, so the unbinned NSBI gain is largest on the k_{2V} (and the low-m_{HH} κ_λ) shape directions. (k_{2V} is now the best-measured "inaccessible" coupling — the 2026 ATLAS+CMS combination gives $k_{2V} \in [0.73, 1.3]$.)
- **Interference impact ambiguity (shared, but worse).** Both have interference, so the finite-difference NP-impact definition is ambiguous; ATLAS replaces it with a local implicit-function-theorem impact [ATLAS p. 90]. κ_λ needs the same fix.

3 Final state and reconstruction

The off-shell 4ℓ final state is *exquisitely clean*: four leptons, fully reconstructed, with tiny energy-scale systematics and no jets/MET/τ/b-tagging in the core. $b\bar{b}\tau\tau$ is the opposite:

- **b-jets** (JES/JER, b-tagging) + **hadronic** τ (τ-ID, τ energy scale) + **MET** (the τ neutrinos). The $\tau\tau$ mass is *not* fully reconstructed — it requires an algorithm (SVfit); shifting τ energy shifts MET *and* SVfit coherently [CMS p. 93].
- **Three channels** with different background compositions and systematics, plus a boosted regime.

Consequence: the observable x for $b\bar{b}\tau\tau$ is softer and carries many more systematic "handles" than 4ℓ.

4 Background handling — the largest qualitative difference

The crux

Off-shell backgrounds are **all MC-based and clean** (one dominant irreducible $q\bar{q}ZZ$ continuum, modelled in simulation, plus EW). $b\bar{b}\tau\tau$ backgrounds are a **zoo, several of them data-driven**, and *data-driven backgrounds have no per-event MC probability density* — which is the object NSBI needs. This is the single hardest conceptual mismatch between $b\bar{b}\tau\tau$ and the off-shell NSBI template.

ATLAS off-shell (Sec. 2.5, Sec. 5). The dominant background is the irreducible $q\bar{q} \to ZZ \to 4\ell$ continuum, which *interferes* with the signal and is carried in the mixture model as a component $\mu_{q\bar{q}ZZ}$ (three floating NPs) constrained *in situ* by a dedicated n_{jets} discriminant in a control region [ATLAS p. 81]. Plus a small EW component. Everything is MC, so every process has a per-event density the NSBI ratios can target.

CMS $b\bar{b}\tau\tau$ (Sec. 7). A heterogeneous set, with the two largest estimated *from data*:

- **QCD multijet — data-driven ABCD** [CMS p. 61,62]. Four regions in (charge × τ-isolation); shape from region C (data minus other-MC), yield from $C \times B/D$. The QCD *normalisation* uncertainty ranges **5–100%** by category (Table 30, [CMS p. 92]); its *shape* uncertainty (alternative B-vs-C templates, symmetrised) is **one of the highest-ranking nuisances** in the most sensitive $\tau_h\tau_h$ channel [CMS p. 91,96,97].
- **Drell–Yan (Z+jets) — MC normalised by 18 data-driven scale factors** [CMS p. 63]. A simultaneous fit of $M(\mu\mu)$ across **18 control regions** (3 b-jet multiplicities × 6 Z-p_{T} bins), giving 18 normalisation SFs (∼0.6–1.7) *and* 18 shape templates (scaled ±) propagated with their covariance [CMS p. 65,94].
- **$t\bar{t}$ — MC shape + data-driven normalisation SF** [CMS p. 79]. A dedicated CR (inverted elliptical (m_{HH}) mass cut, Eq. 22) with $< 7\%$ contamination; COMBINE fit with Barlow–Beeston gives ttSF ≈ 0.91–0.99 [CMS p. 81,82]; the shape is taken from MC (no shape dependence observed).
- **Jet→ τ fakes — data-driven** [CMS p. 94,95]. $> 93\%$ of the events with a jet faking a τ; fake-rate uncertainty per region (barrel 24/13/11%, endcap 28/18/25% by year, Table 33), applied as an event weight.

- **Single Higgs — irreducible, same final state** [CMS p. 82]. ggH, VBFH, VH, $t\bar{t}H$ with $H \to b\bar{b}$ or $\tau\tau$ mimic the signal; modelled from MC. (Note: $t\bar{t}H/VH$ couplings make this a *coupling-dependent* background.)
- **Di-/tri-boson, W+jets, single-t, V+2jets — MC** [CMS p. 82].

Why this is a new, NSBI-specific difficulty

1. **No density for data-driven backgrounds.** QCD (ABCD) and jet$\to\tau$ fakes are estimated as *yields/weights from data control regions*, not as simulated events with a per-event probability. NSBI's per-event likelihood ratio $P_X(x)/P_{\text{ref}}(x)$ is undefined for them. Folding an ABCD/fake estimate into an *unbinned* per-event likelihood is an open problem the off-shell analysis never had to solve (its one big background was MC). Options (all imperfect): treat QCD/fakes only in a binned auxiliary term; build a surrogate density from the control-region data; or restrict NSBI to the MC-modelled part and bin the rest.
2. **The QCD shape systematic is a top-ranked nuisance** [CMS p. 91] — and it is intrinsically *asymmetric/ill-defined* (one alternative template used for both up and down), forcing a symmetrisation hack [CMS p. 96,97]. Carrying that into a per-event morphing is non-trivial.
3. **Single Higgs is an irreducible, coupling-dependent background** in the same final state — it must be modelled *and* morphed consistently with the signal couplings, with its own (correlated) theory systematics.
4. **Many background-normalisation NPs are themselves data-fit results** (18 DY SFs with a covariance, ttSF, QCD B/D) — these are constrained *in situ* in the binned CMS fit; reproducing that constraint inside an unbinned NSBI fit requires the CRs to be represented in the per-event likelihood too.

A workaround: flow-based data-driven backgrounds (ABCDnn)

The "no per-event density" problem is the sharpest objection to NSBI here, but it is *not* a dead end — there is a demonstrated solution. **ABCDnn** (Choi, Lim, Oh, arXiv:2008.03636) generalises the ABCD method with a *neural autoregressive flow* that learns the full multi-dimensional CR$\to$SR transformation, conditioned on the discriminating observables. Where standard ABCD gives only a scalar SR yield ($N_D = N_B N_C/N_A$), ABCDnn gives a full *differential* background distribution $p_{\text{QCD}}(x)$ — which can then be turned into a density ratio $r_{\text{QCD}}(x) = p_{\text{QCD}}(x)/p_{\text{ref}}(x)$ by training a classifier between flow-generated events and the pooled reference, so it enters the NSBI likelihood *on the same footing* as the MC-derived ratios.

The concrete production example (S. An, Dutta/Paulini, CMU PhD thesis 2025, Ch. 5): the CMS *all-hadronic four-top* search, Run-2 ($137\,\text{fb}^{-1}$, 13 TeV) + Run-3 ($62\,\text{fb}^{-1}$, 13.6 TeV), reaching an **observed (expected) significance of** $3.95\,(1.65)\,\sigma$, $\mu = 2.7^{+1.1}_{-0.9}$. Its single most important design choice is a **yield/shape split** of the dominant $t\bar{t}$+jets and QCD backgrounds:

- *Yield* $\leftarrow$ **ExtendedABCD**: standard ABCD in (N_{jets}, N_b) *fails* here (the factorisation breaks at high multiplicity — off by ∼18–19.5% in closure); adding two extra sidebands restores it to ∼1–7% (their $\hat{F}_D = (F_B F_C/F_A)^2 (F_X/F_B F_Y)$, Table 5.1–5.2). The lesson: *get the normalisation from a robust closed-form estimator with enough sidebands, not from the flow.*
- *Shape* $\leftarrow$ **ABCDnn** (a neural autoregressive flow): it learns a transformation that *morphs the $t\bar{t}$ MC into the data-driven $t\bar{t}$+QCD estimate* (target = data − signal − minor bkg per CR), trained by minimising **Maximum Mean Discrepancy** and *conditioned on the CR*

position (N_{jets}, N_b), then applied to $t\bar{t}$ MC in the SR. Notably it never uses QCD MC (too few SR events) — the flow turns pure-$t\bar{t}$ MC into an effective $t\bar{t}$+QCD mixture.

Three training tricks they found necessary (worth copying): **loss-weighted CR sampling** (sample harder CRs more often, $\propto e^{\text{loss}}$, so every region is learned uniformly); **continuous** (N_{jets}, N_b) **conditioning** instead of one-hot labels, so the flow *interpolates smoothly* across CRs — this is what makes the CR→SR *extrapolation* well-behaved; and a **warm-up** (pre-train near identity, RealNVP-style) for stable convergence.

Application to $b\bar{b}\tau\tau$: the **jet→ τ fakes** are the natural target — a fake-factor method is conceptually an ABCD with τ-ID inversion as the second axis, so an ABCDnn conditioned on $(p_{\text{T}}^{\tau}, \eta^{\tau})$ could learn the full fake-τ kinematic shape instead of the current coarse 2D fake-factor binning; the QCD multijet is the other target; and for $t\bar{t}$ the flow can serve as a data-driven shape cross-check. Following the four-top lesson, keep the **yield/shape split**: take the fake-τ/QCD *normalisation* from the existing fake-factor/ABCD machinery (which CMS already trusts, [CMS p. 61,94]) and use ABCDnn only for the differential *shape*, then convert that shape to a per-event ratio $r(x)$ for the likelihood. A fully realised $b\bar{b}\tau\tau$-NSBI would use *ABCDnn shapes (with ABCD/fake-factor yields) for QCD/fakes, MC ratios for the MC-modelled components (Z+jets, $t\bar{t}$, single-H, diboson), and one unbinned NSBI likelihood combining both* — removing the main caveat that NSBI "depends on MC where MC is wrong."

But the generic caveats remain: (i) *extrapolation distance* — the CR→SR map strains for tight SRs far from the sidebands in tagger/isolation space (multiple overlapping CRs validate it but cost stats); (ii) *uncertainty propagation* — the four-top analysis does *not* bootstrap the flow; it splits the uncertainty into three data-driven nuisances (thesis Sec. 6.1, Table 6.1): the *finite-CR-statistics* piece is a lnN normalisation NP propagated through the ABCD formula ("statistics of extended-ABCD," 4–26%); the *normalisation-modelling* piece is a lnN NP from a validation-region (VR) data-vs-prediction closure (weighted-mean deviation $1 - \langle f \rangle$ + weighted RMS of $f = N_{\text{data}}/N_{\text{pred}}$, 6–44%); and the *ABCDnn shape* uncertainty is a shape NP implemented as a *linear shift of the discriminant*, with the shift size chosen by *minimising a quantitative data/pred-agreement metric* in the VR (floor 1%, correlated across H_T, uncorrelated across years). The whole uncertainty-*estimation* procedure is itself validated in a dedicated orthogonal "closure-test region" that checks pred+syst covers truth in the SR. *Caveat for NSBI:* this VR-closure shape nuisance ("shift the discriminant") is a *binned-fit* construct; in an unbinned NSBI it becomes yet another per-event morphing nuisance on the ratio $r(x)$ — i.e. the same learned-morphing problem, now applied to the data-driven component; (iii) *signal contamination in CRs* — if signal leaks into the control region the flow learns "background + signal" and biases the SR low, requiring iterative CR definitions with signal subtraction (this caveat is *mild* for $b\bar{b}\tau\tau$: the HH signal is tiny in the fake-τ/QCD control regions, just as the SM four-top signal was negligible in its CRs); (iv) *no theory handle* — the flow gives a shape but no ME+PS+PDF decomposition, so generator-level theory uncertainties on the background cannot be propagated and are replaced by a data-driven CR→SR closure systematic.

The one caveat specific to using ABCDnn *for NSBI* (the real gap). In the four-top thesis the flow models only a *low-dimensional* target — the event-level BDT discriminant plus H_T (two variables) — because the downstream fit is a *binned template* fit of that discriminant. NSBI needs the opposite: the *full per-event density* over the whole feature vector, so the data-driven shape can become a per-event ratio $r(x)$. So the four-top result proves flows can deliver a data-driven background *shape of a summary observable*; extending that to the full-dimensional per-event density an unbinned NSBI consumes is a genuine step beyond what is demonstrated — a higher-dimensional flow, harder to train and validate. The conservative intermediate is to let ABCDnn

produce the data-driven shape of the *NSBI discriminant* (the per-event log-ratio summed over components), i.e. treat QCD/fakes as a binned component of an otherwise-unbinned likelihood — exactly the "keep QCD binned, NSBI-fy the MC components" compromise.
Net: ABCDnn downgrades the data-driven-background problem from "open" to "solved-with-caveats" for a binned *discriminant shape; the fully-unbinned version is a worthwhile but unproven sub-project for a $b\bar{b}\tau\tau$-NSBI pilot.*

5 Systematics handling

What is genuinely shared (equal for both, and equally fallible): the *sources and* $\pm 1\sigma$ *sizes* of the NPs; the *object-level propagation* (shift the object and all dependents — e.g. $\tau \Rightarrow$ MET$\Rightarrow$SVfit — and recompute the discriminant, [CMS p. 93]); the *factorisation over NPs* and $\pm 1\sigma$ *interpolation*; *Gaussian-constrained profiling*; and the κ_λ*-dependent theory* treatment. ATLAS builds the per-event systematic weight as $P_X(x,\alpha) = w_X(x,\alpha)\,P_X(x,0)$ with $w_X(x,\alpha) = \prod_m w_X(x,\alpha_m)$, each single-NP weight interpolated poly-exp between $\pm 1\sigma$ classifier estimates [ATLAS p. 83,84]. CMS does the per-*bin* analogue (re-run the DNN on varied inputs $\to$ template).

What is harder for $\kappa_\lambda/b\bar{b}\tau\tau$:

1. **κ_λ–systematic shape degeneracy.** The dominant experimental systematics (JES, τES, MET, signal parton shower) deform *the same m_{HH} shape* that carries κ_λ. In 4ℓ the big systematics (theory) are largely *separable* from the signal shape; in $b\bar{b}\tau\tau$ they are not. Decorrelation is therefore impossible (it would erase the signal), forcing the "model-it" route — so the accuracy of the morphing $\Delta(x|\alpha)$ *directly sets* the κ_λ uncertainty. (A Fisher-information toy gives, schematically, a $\sim$13$\times$ systematic inflation from m_{HH} alone collapsing to $\sim$1.3$\times$ with the full feature set — the degeneracy is broken by dimensionality, not avoided.)

2. **An order of magnitude more, and larger, experimental NPs** [CMS p. 90–97]: JES split into a reduced set of **11 sources** with a year-correlation scheme [CMS p. 95]; τ**ES** p_{T}-dependent with 4 decay-mode components [CMS p. 94]; ***b*-tag** 5 components (hf, lf, hfstats, lfstats, cferr) [CMS p. 96]; **DeepTau** vsJet (5 p_{T} bins) + vsEle (2 η bins) [CMS p. 95,96]; trigger SFs, JER, PU-jet-ID, L1 prefiring, $e/\mu \to \tau$ fake ES. The off-shell analysis has comparatively few, small (lepton) experimental NPs. Each NP is another per-event weight to learn ($\sim$2000 NNs already in off-shell = 2$\times$NP$\times$process; $b\bar{b}\tau\tau$ grows this).

3. **Data-driven background systematics that don't fit the density-ratio picture**: the QCD norm (5–100%) and high-ranking QCD shape [CMS p. 91,92,96,97], the 18 DY SF shape/norm templates [CMS p. 94], the jet$\to\tau$ fake-rate uncertainties (11–28%, Table 33) [CMS p. 95], the VBF dipole-recoil PS uncertainty (20–70%, Table 32) [CMS p. 93]. These are template/weight constructs, not deformations of an MC density.

4. **κ_λ-dependent theory uncertainty (POI$\times$systematic entanglement).** The HH scale/PDF/m_t uncertainties are folded into a normalisation NP `THU_HH` $= +4\%/-18\%$ at the SM point that becomes a κ_λ-*dependent* uncertainty across the scan [CMS p. 91]. The off-shell μ does not entangle with its theory uncertainty this way. Capturing it requires non-factorisable (POI-coupled) morphing.

5. **Factorisation is more likely to break.** $w_X = \prod_m w_X(\alpha_m)$ assumes NPs are independent sources [ATLAS p. 83]; but JES and τES *both* feed m_{HH} $(= m(b\bar{b}+\tau\tau))$, so a genuine cross-term exists in $b\bar{b}\tau\tau$ that is benign in clean 4ℓ.

6. **MC-statistical uncertainty of the learned ratios.** ATLAS propagates it as one NP via the NN-ensemble spread [ATLAS p. 86,87]; CMS uses Barlow–Beeston per bin. With far smaller *systematic-variation* samples in $b\bar{b}\tau\tau$, this term is more important (the nominal MC, ≈500k–750k/κ_λ-point, is ample for training; the binding constraint is the least-populated $\pm 1\sigma$ sample).

Two systematics that NSBI actually handles *better* than templates. Worth stating the upside too: (a) *parton-shower / generator* uncertainties (Pythia vs. Herwig, ISR/FSR) are a natural fit — a **CARL/DCTR**-style reweighting network learns the per-event weight between alternative showers, which is then profiled as a nuisance; this is strictly more powerful than the current "shape envelope over generator variations" and is the in-likelihood cousin of the RS3L robust-representation idea. (b) b-tag / τ-ID scale factors can be *absorbed* into the density ratio and calibrated in situ against a standard candle (the $VZ(c\bar{c})/Z \to b\bar{b}$ peak trick), turning the POG SF program into an external cross-check rather than the primary constraint. So the systematics ledger is not all cost: *the morphing-heavy ones (JES/τES on m_{HH}) are the burden; the modelling ones (shower, flavour-tagging) are where NSBI's per-event treatment is an asset.*

6 Statistics and inference

- **Data signal yield is tiny and intrinsic.** $\sigma \times \mathcal{B} \sim 2.5$ fb $\Rightarrow \mathcal{O}(10\text{–}100)$ selected signal events (Run-2 to HL-LHC), under large backgrounds — vs the off-shell tail at ∼15% of σ. This is unchanged by MC and keeps the measurement systematics-comparable. (It is also the regime where NSBI's per-event information helps *most*.)
- **MC/training stats are ample** (≈500k–750k per κ_λ point), so the morphing/ratio NNs are well-trained — the "under-determined morphing" worry applies to the *systematic-variation* samples, not the nominal.
- **Validation & coverage.** CMS inherits decades-validated COMBINE (asymptotics + toys + Barlow–Beeston). NSBI needs a custom JAX/IMINUIT fit with an explicit Neyman construction and ratio-quality diagnostics (C2ST-like) built and validated from scratch [ATLAS p. 64,89].

7 Synthesis: could the NSBI κ_λ result be *worse* than the CMS fit?

Reading what CMS *actually* does removes a strawman: CMS does **not** bin m_{HH}; it bins a **multivariate DNN score** (shape systematics propagated by re-running the DNN on varied inputs, [CMS p. 93]). So CMS already carries the same multi-dimensional systematic information NSBI does — **NSBI is not "exposed to more systematics" than the real CMS baseline.** Against CMS-as-done, the comparison reduces to:

- **What NSBI buys:** *unbinning* (no information lost to binning the score) and a κ_λ-*optimal* per-event ratio (CMS's DNN is one fixed discriminant, not optimal at every κ_λ). These are the off-shell ∼2× / FAIR-Universe ∼20% gains.
- **What NSBI costs:** a *learned* morphing layer (NSBI must *train* the per-event systematic weight; CMS *counts* its re-run-DNN template, exact up to MC stat) and the *custom coverage validation.* The danger of "worse than CMS" is the *learned layer being mis-calibrated / under-validated* — producing a tighter-but-mis-covering interval — not a heavier systematics burden.

- **The genuinely new $b\bar{b}\tau\tau$ problem is the data-driven backgrounds** (Sec. 4): if QCD/-fakes cannot be given a faithful per-event density, the unbinned NSBI may have to bin them anyway, partially eroding its advantage in exactly the most sensitive channel.

Protections (so the worst case is "≈ CMS," not "worse and unknown"): honest morphing uncertainty (GP-style); explicit coverage validation; and **binning the NSBI discriminant as a built-in cross-check** (the binned fit is a coarsening of NSBI, so a regression is detectable and you fall back).

8 Summary of new challenges

	ATLAS off-shell 4ℓ	**κ_λ in HH→ $b\bar{b}\tau\tau$ (new challenge)**
POI	$\mu_{\text{off-shell}}$ (1 POI)	κ_λ ($+k_{2V}, k_V$, EFT); shape in m_{HH}
Final state	4 leptons, fully reco, clean	b-jets+τ_h+MET; SVfit; 3 channels
Backgrounds	one MC $q\bar{q}ZZ$ (interfering), CR-constrained	QCD (ABCD), DY (18-CR SFs), $t\bar{t}$ (CR-SF), **jet→ τ fakes**, single-H, diboson
Data-driven bkg in NSBI	none needed	**QCD & fakes have no per-event density**
Experimental systematics	few, small (leptons)	many, large (JES×11, τES, b-tag×5, DeepTau, ...)
Syst. vs signal shape	largely separable	**JES/τES deform m_{HH} (degenerate with κ_λ)**
Theory–POI coupling	none	**κ_λ-dependent** `THU_HH` $+4/-18\%$
Factorisation of NPs	safe	JES×τES cross-term in m_{HH}
Signal yield	∼15% of σ	$\mathcal{O}$(10–100) events (fb-scale)
MC training stats	—	ample (≈500–750k/κ_λ); watch $\pm1\sigma$ samples
Stats framework	custom (built)	must port the custom fit; CMS has COMBINE

Bottom line

The NSBI *method* ports cleanly from off-shell to κ_λ. The new challenges are in the *inputs*, and rank roughly: (1) **data-driven backgrounds** (QCD, jet→ τ fakes) with no per-event density — the most awkward fit with NSBI, but *solved-with-caveats* by flow-based estimation (**ABCDnn**, Sec. 4); (2) the **κ_λ–systematic shape degeneracy** (JES/τES on m_{HH}) plus a much larger experimental-systematics surface, which makes *morphing accuracy + honesty* the limiting factor; (3) the **κ_λ-dependent theory** and **factorisation cross-terms** (non-factorisable morphing); (4) **irreducible single-Higgs**. None are NSBI-specific *except* the data-driven-background mismatch and the learned-morphing layer; the rest are equally CMS's problems, handled with the same NPs. Two scope caveats: NSBI's gain is in the *κ_λ/k_{2V} parameter constraint* (shape directions), *not* the SM-HH observation significance (rate-saturated, luminosity-set); and the genuine per-event edge is on *shape/correlated-operator* directions, not the rate-broken $\kappa_t \leftrightarrow \kappa_\lambda$ / $k_V \leftrightarrow k_{2V}$ ones.

References

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[2] CMS Collaboration (C. Amendola *et al.*), *Search for Higgs boson pairs produced through gluon and vector boson fusion in the $b\bar{b}\tau\tau$ final state with Run-II data*, CMS AN-18-121 (2022). Key: signal reweighting Sec. 3.1 (p. 6); background estimation Sec. 7 (pp. 61–85; QCD ABCD pp. 61–62, DY 18 CRs pp. 63–78, $t\bar{t}$ CR-SF pp. 79–82); systematics Sec. 9 (pp. 90–97).

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[5] ATLAS+CMS HH combination, arXiv:2602.23991 / CMS-PAS-HIG-25-014 (early 2026): $k_{2V} \in [0.73, 1.3]$, $\kappa_\lambda \in [-0.71, 6.1]$, $\mu_{HH} < 2.5$ (95% CL).